

Tools for Scientific Discovery

Engineering Design

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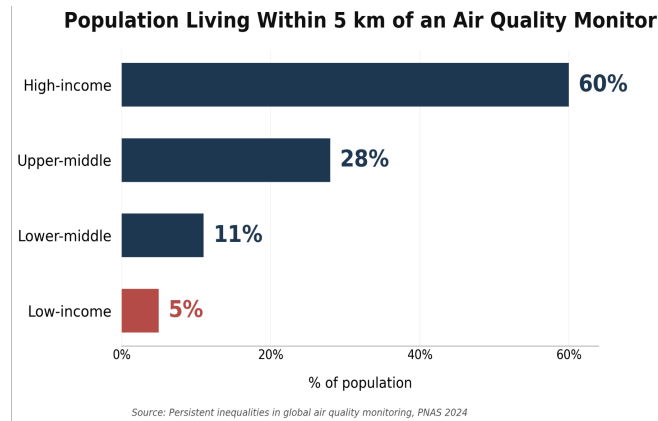
2026

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Identification and Definition of the Problem

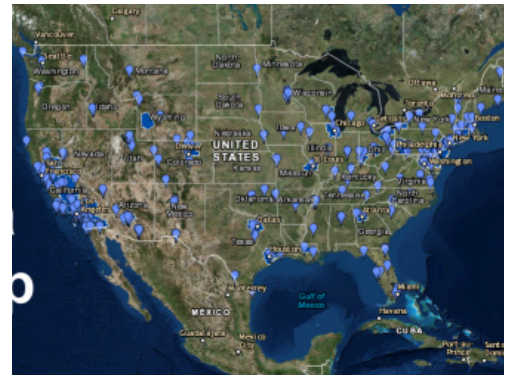
Professional atmospheric monitoring instruments cost between **\$15,000 and \$70,000 per unit**, limiting air-quality observation to well-resourced institutions and leaving many vulnerable communities without access to environmental data. This is not just a technology gap, but a failure in how scientific tools of discovery are designed and distributed.



This scarcity is unequal. Smith et al. found that **60% of people in high-income countries live within 5 km of a monitoring station, compared to only 5% in low-income countries**, a 12× disparity in monitoring access. The same study found that 95% of the world's 33,984 monitoring stations are located in Asia, Europe, or North America, while less than 4% are located in Africa or South America.

Uneven EPA monitoring coverage leaves **many communities without accurate local air-quality data**

Source: EPA AirNow Interactive Map, accessed 2026



Cost and Distribution of Existing Instrumentation

Federal Reference Method (FRM) monitors define the regulatory standard for air quality measurement, but their cost places them beyond the reach of community-level deployment. The Thermo Scientific TEOM 1405 particulate monitor and Horiba APMA-370 carbon monoxide analyzer cost **\$20,000-\$35,000** per unit (Thermo Fisher Scientific; Horiba). At the same time, the EPA Air Quality System (AQS) maintains fewer than **4,000 fixed monitoring sites** across the contiguous United States (EPA AQS, 2024), leaving many rural, wildfire-adjacent, and industrial communities without accurate local data.

Existing Community-Scale Solutions

Existing platforms revealed **important gaps**. PurpleAir provides **low-cost distributed PM2.5 sensors**, but **lacks gas sensing** and a **full calibration framework**. OpenAQ and AirNow rely on **sparse government monitoring**. None fully combine **low cost, multi-pollutant sensing**, and **community-scale architecture**.

Information Gathering - Engineering The Tools of Scientific Discovery

To understand the need for affordable atmospheric monitoring, we researched the consequences and health impacts of this instrumentation gap, and the history of air-quality measurement. Across these areas, a conclusion emerged that the **limiting factor is not the ability** to measure pollution, **but access to the tools** needed to do it.

The Consequences of the Instrumentation Gap

1. **Undetected Health Hazards:** Without local monitoring instruments, dangerous concentration events from CO, NO₂, and PM2.5 pollutants linked to severe respiratory and cardiovascular illness go undetected, producing gaps in the scientific record that prevent meaningful response.
2. **Inability to Advocate for Change:** Scientific discovery drives policy. Communities that lack the instrumentation to measure local pollution cannot generate the quantitative, reproducible data needed to compel regulatory action or hold polluters accountable.
3. **Exclusion from Scientific Participation:** Citizen science expands environmental research, but participation requires accessible instrumentation. Without engineered tools designed for affordability and ease of deployment, entire communities are excluded from contributing to discovery.

Health and Environmental Impact

PM2.5 can **penetrate lung tissue and the bloodstream**, with documented links to **respiratory and cardiovascular disease** (Newby et al., 2015; Du et al., 2016). The World Health Organization estimates that **ambient air pollution** causes approximately **4.2 million premature deaths annually** (WHO, 2024). Elevated ground-level ozone causes over \$500 million in annual U.S. agricultural losses (USDA, 2014), and contributes to broader ecosystem damage on soil chemistry and biodiversity.

Historical Perspective

Historically, modern air monitoring expanded after the **1952 Great Smog of London** and the **1970 U.S. Clean Air Act**. However, the instruments developed for regulatory accuracy were designed for **institutional use, not broad public access**. As wildfire smoke, urban growth, and industrial emissions increasingly affect unmonitored areas, the need for **affordable sensing** has become more urgent.

Solving this problem requires engineering that **reduces cost and complexity without compromising scientific usefulness and public access for community-scale monitoring**. Thus, the task is to create a **deployable alternative** to expensive reference monitors that **expands who can measure** local air quality **and participate in scientific discovery**.

Possible Solution #1: Wearable Air Quality Sensor Module

Traditional regulatory air-quality stations produce fixed data points that are often used to represent broad regions. However, the pollution a person actually inhales depends on where they travel throughout the day. A wearable air-quality sensor module would address this gap by generating a continuous, personal exposure profile as the user moves through commutes, schools, workplaces, and homes.

Scientific Basis

- **Particulate Detection:** A compact optical particle sensor would measure PM1.0, PM2.5, and PM10 using laser-scattering technology, providing real-time data on particles strongly linked to respiratory and cardiovascular health risks.
- **Gas Detection Array:** Electrochemical and MOS sensors target CO, NO₂, O₃, and VOCs, with temperature and humidity co-readings for environmental correction.
- **Microcontroller Core:** An ESP32 manages sensor polling, onboard flash storage, and BLE transmission to a paired smartphone while using under 500 mW of power.

Engineering Feasibility

- **Form Factor & Enclosure:** Designed as a clip-on or wristband attachment with a miniaturized air inlet. Fabricated in PETG or ABS via FDM 3D printing for prototyping, injection molding for production.
- **Power Management:** Dynamic firmware scales sampling rate and duty cycle by activity state. A 1000–2000 mAh LiPo cell provides full-day operation at 1 Hz intervals.
- **Data Pipeline:** BLE streams readings to a companion app for real-time visualization, while onboard flash buffers data, ensuring no loss in low-connectivity environments.

Building on What Exists - And Going Further

- The wearability adds mobility, allowing people to measure air quality as a lived experience rather than at a fixed location.
- Aggregated across many users, this generates pollution maps tied to human movement, answering questions about commute routes and workplaces that fixed monitors may miss.

Public Health & Environmental Benefits

- Enables community-based participatory research, generating movement-based exposure maps that could provide policymakers the evidence needed to target interventions.
- Empowers individuals with real-time risk awareness. Users can avoid chronic exposure hotspots, reducing inhalation of PM2.5 and NO₂, both linked to asthma, cardiovascular disease, and premature mortality.
- Chronic exposure data can support analyses of exposure-disease relationships in populations too small or dispersed for traditional epidemiological studies.

Limitation: It measures personal exposure more effectively than continuous neighborhood-wide conditions.

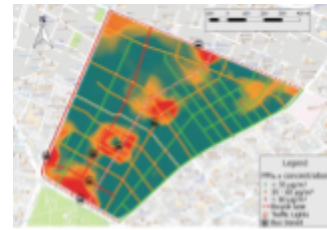


Figure 1. PM levels vary across a city, showing why mobile exposure tracking can be useful

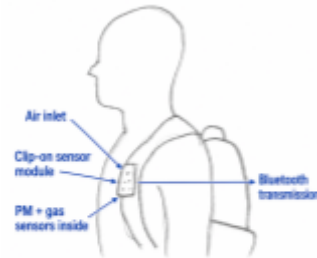


Figure 2. Concept sketch of a wearable air-quality sensor module for personal exposure monitoring.

Possible Solution #2: Fixed Mesh Network

A fixed-node mesh network distributes inexpensive sensor units across a geographic area, with each node sampling local atmospheric conditions and relaying data through a wireless network. This architecture trades per-node precision for spatial density, generating a pollution gradient map that reveals micro-scale variation that isolated instruments often miss.

Scientific Basis

- **Particulate Detection:** Each node integrates a laser-scatter optical particle counter for PM1.0, PM2.5, and PM10. **Cross-node calibration** helps reduce sensor drift by comparing nearby nodes and identifying inconsistent readings.
- **Gas Detection Array:** Low-power MOS sensors target CO, NO₂, and O₃, with onboard temperature and humidity sensors providing environmental correction for more consistent readings.
- **Microcontroller Core & Mesh Communication:** A microcontroller manages sensing and mesh networking. ESP-MESH supports peer-to-peer routing between nodes, while LoRaWAN can extend range through gateway-based communication. Gateway nodes upload timestamped packets over Wi-Fi or cellular to a cloud database.



Figure 1. Mesh network concept: sensor nodes relay data to a gateway before cloud upload.

Engineering Feasibility

- **Enclosure & Deployment:** Weather-resistant enclosures can mount to existing infrastructure, such as lamp posts, fence lines, building facades. Optional solar panels and LiFePO₄ batteries could support off-grid operation in locations without reliable power.
- **Scalability:** A low-cost sensor node enables community deployment at a fraction of the cost of reference-grade monitors. Mesh routing can also improve reliability by allowing nearby nodes to relay data if one communication path fails.
- **Data Pipeline:** Cloud backend aggregates sensor streams, applies anomaly detection, and serves a public dashboard with historical trend visualization and downloadable datasets for academic, community, or advocacy use.

Building on What Exists - And Going Further

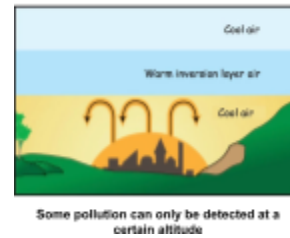
- **Proven concept:** Existing community sensor networks such as PurpleAir show that low-cost monitoring can produce scientifically useful local air-quality data.
- **Infrastructure independence:** Solar power and long-range wireless communication allow deployment beyond homes or areas with reliable Wi-Fi.
- **Network reliability:** Mesh or gateway-based routing helps maintain data flow if individual nodes lose connection.

Public Health & Environmental Benefits

- **Block-level monitoring:** Dense sensor placement can capture local events such as vehicle idling, industrial emissions, and wildfire smoke that single stations may miss.
- **Public data access:** Anonymized data feeds could support community advocacy, environmental research, and studies connecting local exposure patterns with health outcomes.

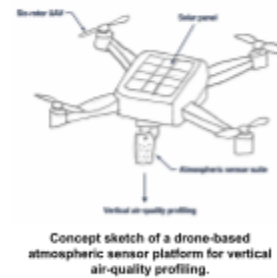
Possible Solution #3: Solar-Charged Drone with Atmospheric Sensor Suite

Fixed sensors and wearables measure air quality near ground level, but they cannot easily capture how pollution changes with altitude. Duclaux et al. showed that ground monitors can miss vertical pollution patterns, including high-altitude ozone transport. A drone-based sensor suite would act as a mobile vertical profiler, collecting data on thermal inversions, plume movement, and pollutant transport.



Scientific Basis

- **Particulate Detection:** An OPC-N3 optical particle counter provides real-time PM1, PM2.5, and PM10 readings during GPS-guided flight.
- **Gas Detection Array:** An electrochemical array targets CO, NO₂, SO₂, and O₃. A ventilated nacelle improves airflow across the sensors, while correction models account for drone-induced airflow effects.
- **Autopilot & Data Systems:** ArduPilot or PX4 flight controllers execute pre-programmed missions with GPS waypoint navigation, logging sensor data in GIS-compatible formats.



Engineering Feasibility

- **Platform & Power:** A six-rotor UAV provides 20-40 minutes of flight time per charge, while a solar charging station could support multiple daily flights at around \$1,200 total cost.
- **Autonomy & Safety:** Geofencing, return-to-home failsafes, and trained operation would support safe deployment under FAA Part 107 guidelines.
- **Data Pipeline:** Post-flight logs uploaded via Wi-Fi to a cloud platform, where interpolation tools generate 3D pollution maps for research, emergency response, and community advocacy.

Building on What Exists - And Going Further

- **3D Atmospheric Profiling:** Lidar and research aircraft have shown that vertical measurements reveal pollution movement missed by ground stations. A drone makes this capability more repeatable and accessible.
- **Repeatable precision:** GPS waypoint navigation enables identical routes across days or months, building time-series datasets that track pollution changes across seasons, emergencies, or policy interventions.

Public Health & Environmental Benefits

- **Vertical Pollution Detection:** Drone profiling can identify temperature inversions that trap pollutants near ground, supporting more accurate public health advisories.
- **Targeted Emergency Use:** Deployment after industrial incidents, wildfires, or acute pollution events could give responders 3D awareness that fixed sensors and wearables cannot provide.

Comparison of Solutions

	Weight	Solution #1 Wearable Sensor Module	Solution #2 Fixed Mesh Network	Solution #3 Autonomous Drone
Affordability <i>Cost per unit</i>	25% - An unaffordable system fails the mission regardless of technical merit.	Low ~\$50-80 per unit; no fixed infrastructure needed; rechargeable battery Rating: 4/5	Lowest ~\$40 per node; scalable across many locations; optional solar support; Rating: 4/5	High ~\$1,200 full system; FAA Part 107 compliance and trained operation cost Rating: 2/5
Spatial coverage <i>Geographic reach</i>	25% - Neighborhood-scale coverage is the network's core value proposition.	Personal only Tracks individual exposure along personal routes; biased towards high population density Rating: 2/5	Block-level Dense fixed grid produces continuous, neighborhood-scale pollution maps Rating: 5/5	3D vertical Altitude profiling captures inversion layers; limited flight time per charge Rating: 4/5
Deployability <i>Ease of setup</i>	20% - Complex setup creates adoption barriers in under-resourced communities.	Easy Clip-on form factor; no installation; requires paired smartphone Rating: 4/5	Easiest Mounts to lamp posts and fences; self-healing mesh; minimal user intervention after installation Rating: 5/5	Complex FAA registration required; needs trained operators for each flight Rating: 2/5
Scientific value <i>Data quality</i>	20% - Data quality is secondary to cost and coverage.	Novel personal data Exposure profiles along daily routes; not suitable for community-scale analysis Rating: 3/5	Strong Continuous long-term local datasets; public API access for researchers Rating: 5/5	Unique 3D data Trajectory mapping impossible with ground systems; limited temporal resolution Rating: 5/5
Key weakness <i>Primary limitation</i>	10% - All three solutions share addressable limitations. Architecture is the primary differentiator.	No community coverage Data meaningful only at individual level; cannot identify neighborhood-wide hazards Rating: 3/5	Total-system cost Total system cost scales with node count Rating: 4/5	Not continuous Limited by flight time, weather, FAA rules, and trained operation. Rating: 2/5
Total	100%	3.20/5	4.65/5 - Chosen	3.10/5

Selected Solution: O.A.S.I.S. Fixed Mesh Air-Quality Network

O.A.S.I.S. (Open-Access Atmospheric Sensing & Information System) is a fixed community air-quality monitoring network built on an Internet of Things (IoT) architecture, designed to deliver affordable, real-time, local atmospheric data at the neighborhood scale. Built on an ESP32 microcontroller with a hand-etched custom PCB, each node integrates a PMS5003 laser-scatter sensor for PM2.5 and PM10, a MICS6814 metal-oxide array for NO₂, CO, and NH₃, and a BME280 environmental sensor for temperature, humidity, and atmospheric pressure. This multi-pollutant, context-aware sensor helps readings be interpreted with environmental context for around \$40 per node.

Data streams continuously via Wi-Fi to an InfluxDB time-series cloud database, where readings are processed against EPA Air Quality Index thresholds and surfaced through a custom Android mobile application built in Kotlin and Java. The app delivers color-coded AQI readings, threshold-based push alerts, and personalized health guidance based on user profiles including age and respiratory conditions, translating raw sensor data into protective decisions without requiring any technical literacy.

While O.A.S.I.S. is designed for a distributed mesh architecture, the current prototype validates the core node hardware and data pipeline in a star configuration, where each node connects independently to the cloud. Full mesh networking via LoRaWAN is a planned next-generation feature.

O.A.S.I.S. addresses the grand challenge of engineering the tools of scientific discovery by democratizing environmental monitoring. Professional reference instruments cost \$15,000–\$70,000 per unit, restricting air quality observation to well-funded institutions. At approximately \$40 per node, O.A.S.I.S. enables the deployment density those instruments cannot reach, capturing block-level pollution gradients, transient emission events, and community-specific exposure patterns that single government stations may miss. Time-stamped, geospatially tagged data can be exported in standard formats, allowing residents, students, and researchers to contribute to epidemiological and environmental justice work that has historically required institutional access.



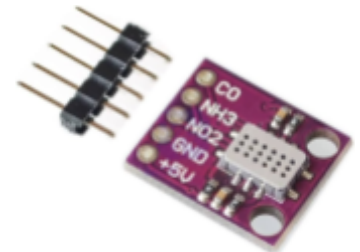
3D Render of the Assembled Module

Components of O.A.S.I.S.

Multi-Gas Pollutant Sensor Array

The core of O.A.S.I.S. includes a comprehensive set of sensors designed to detect the most health-relevant atmospheric pollutants:

- **Nitrogen Dioxide (NO₂) Sensor:** Detects this traffic and industrial pollutant which contributes to respiratory irritation and asthma exacerbation. NO₂ levels are critical indicators of combustion-related air quality issues.
- **Carbon Monoxide (CO) Sensor:** Monitors this colorless, odorless gas produced by incomplete combustion in vehicles, furnaces, and wildfires. CO can be immediately dangerous at high concentrations and is a key indicator of combustion safety.
- **Ammonia (NH₃) Sensor:** Tracks agricultural and industrial emissions that contribute to particulate matter formation and ecosystem disruption. NH₃ monitoring is particularly important near farms and industrial facilities.
- **Particulate Matter Sensor (PM2.5 and PM10):** Uses laser scattering technology to detect fine and coarse particles that penetrate deep into lungs and bloodstream. PM is among the most health-damaging pollutants and a critical wildfire smoke indicator.



MICS6814 detects CO, NO₂, & NH₃



PMS5003 detects particulate matter

Environmental Context Sensor

This integrated sensor module continuously measures three critical atmospheric parameters that dramatically influence pollutant behavior and human health risk:

- **Temperature:** Affects chemical reaction rates, pollutant dispersion, and human vulnerability to air quality impacts. Higher temperatures can increase ground-level ozone formation and worsen health effects.
- **Humidity:** Influences particulate matter size and behavior, sensor accuracy, and the formation of secondary pollutants. High humidity can cause particles to swell and become more visible as haze.
- **Atmospheric Pressure:** Indicates weather pattern changes that affect pollutant accumulation and dispersion. Low pressure systems can trap pollutants near the ground, while high pressure often brings clearer conditions.



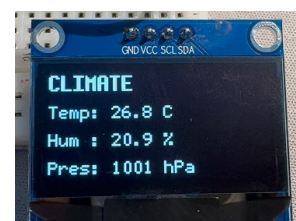
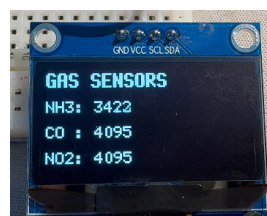
BME280 detects environmental context

By integrating these environmental measurements alongside pollutant detection, O.A.S.I.S. provides additional atmospheric context that supports more accurate interpretation and meaningful health guidance.

Data Transmission and IoT Connectivity

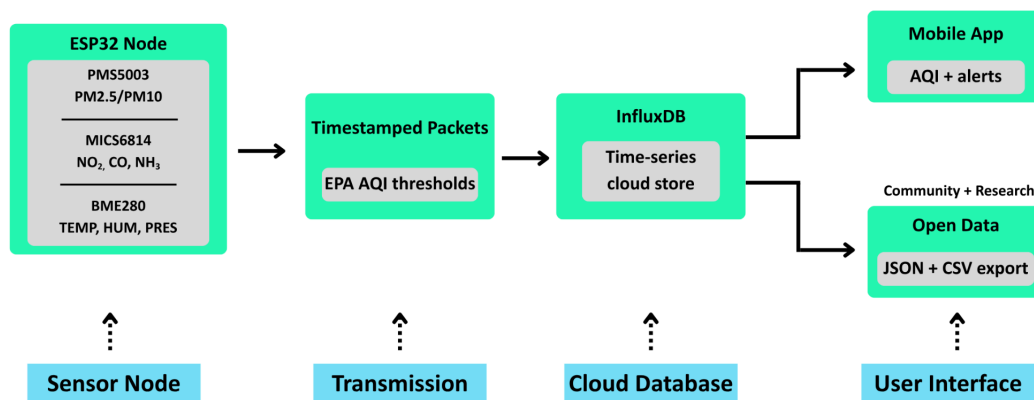
O.A.S.I.S. employs robust wireless connectivity to ensure real-time data availability:

- **Wi-Fi Primary Connection:** Nodes connect to local wireless networks for high-bandwidth, low-cost data transmission in areas with reliable Wi-Fi coverage.
- **InfluxDB Cloud Time-Series Database:** All measurements are transmitted to this specialized database optimized for handling continuous sensor streams. The database timestamps each reading and enables efficient historical queries and trend analysis.
- **Real-Time Alert Engine:** The system continuously analyzes incoming data against health-based thresholds (EPA Air Quality Index standards), or adjustable threshold values and triggers immediate alerts when hazardous conditions are detected.
- **Cloud-Based Processing:** Data aggregation, quality control checks, and spatial analysis occur in the cloud, reducing onboard computational requirements and enabling sophisticated analytics across entire monitoring networks while optimizing power usage.



OLED display showing live sensor readings

O.A.S.I.S. SYSTEM ARCHITECTURE



User Interfaces and Community Access

O.A.S.I.S. delivers environmental data through a platform designed for different user needs and technical literacy levels.

Mobile Application:

- Push notifications when air quality reaches unhealthy levels
- Personalized health recommendations based on user profiles (age, respiratory conditions, activity level)
- Community reporting tool for documenting visible pollution sources or unusual conditions

Data Exports:

- Open data access for researchers, app developers, and community organizations
- Standard data formats (JSON, CSV) for easy integration with other systems
- Real-time and historical data endpoints
- Documentation for third-party application development

Power System

O.A.S.I.S. stations are designed for low-maintenance continuous operation once deployed. The production design draws power directly from a home's existing AC electrical supply, stepping down to the ESP32's 5V requirement, eliminating battery replacement and enabling 24/7 monitoring.

- **Grid Power (Production Design):** Stations connect to standard AC power, optionally paired with a small battery backup to maintain operation during brief outages.
- **9V Battery (Prototype & Field Testing):** The Current prototype uses a 9V battery with a voltage regulator, enabling the portable, untethered operation needed for field validation testing, deploying the same unit at roadside, residential, and school campus sites without requiring electrical access at each location.
 - **Voltage regulator:** In order to meet voltage regulations of the ESP32, a voltage regulator is used to step down the 9V provided by the battery to 5V, providing a safe, clean supply of power to the microcontroller.

Weatherproof Enclosure

The protective housing ensures long-term outdoor reliability:

- **UV-Resistant Material:** 3D-printed PETG enclosure with side-mounted angled louvered vents and sliding access panels; weather-resistant, validated against simulated rain testing.

- **Ventilated Sensor Chamber:** Air flows naturally through the sensor compartment via particle-filtered vents and rain shields, ensuring representative atmospheric sampling without water intrusion.
- **Mounting Hardware:** Versatile back bracket enables installation on poles, building walls, or freestanding posts with secure, stable positioning.

Bill of Materials

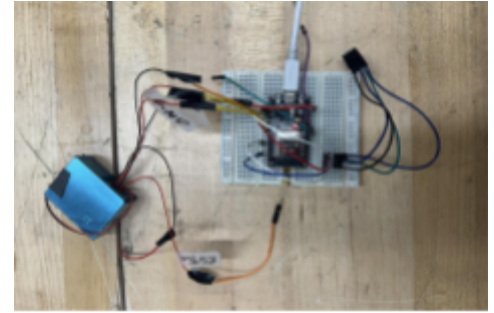
This bill of materials reflects the production design of O.A.S.I.S. and therefore omits prototype-only components (9V battery, voltage regulator, rocker switch). Component pricing is sourced from AliExpress and assumes small bulk ordering (10+ units), which is consistent with a community-scale deployment scenario.

#	Component	Purpose	Est. Unit Cost
1	ESP32 WROOM-32 Dev Board	Microcontroller, Wi-Fi, Bluetooth	\$4.50
2	PMS5003	PM2.5 / PM10 laser scatter sensor	\$17.00
3	MICS6814 Breakout	NO ₂ , CO, NH ₃ metal oxide array	\$8.00
4	BME280 Breakout	Temperature, humidity, pressure	\$5.00
5	SSD1306 OLED Display (0.96")	On-unit status readout	\$3.00
6	Copper-clad board + ferric chloride	Hand-etched custom PCB	\$2.00
7	PETG filament (per-unit portion)	3D printed enclosure	\$2.50
8	Misc. (resistors, headers, wires, screws)	PCB passives and assembly hardware	\$2.00
Total Price			\$44.00

Prototypes

Prototype V1 (Breadboard Development)

Our initial prototype focused on proving core sensor functionality and data transmission capabilities. This version used a breadboard platform with an Esp32 microcontroller connected to individual sensor modules. The sensors were mounted on a simple frame with minimal weather shielding created from a plastic container with ventilation holes. Power came from a wall adapter or laptop output during testing.



Key Learnings from V1:

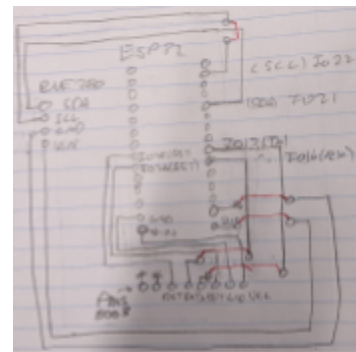
- Sensors provided reasonable readings but showed drift over 24+ hours without calibration
- Wi-Fi transmission to InfluxDB worked reliably, but required manual network configuration
- Basic enclosure was insufficient for outdoor deployment-moisture condensation occurred overnight



First enclosure prototype

Prototype V2 (Integrated PCB Design)

Based on V1 learnings, we created a custom printed circuit board (PCB) that integrated all sensors, the microcontroller, and power management into a compact unit. This version added the BME280 sensor for environmental context data. We upgraded to an isogridded project box with better positioned vents that allowed air exchange while accommodating for the addition of the PCB and battery.



Sketch of PCB layout

Key Improvements in V2:

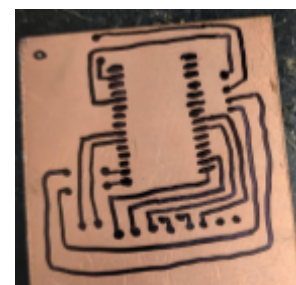
- PCB integration reduced wiring complexity and improved reliability
- Battery system enabled 48-hour mobile operation



Second enclosure prototype

Remaining Limitations in V2:

- Dashboard required technical knowledge to interpret raw sensor values
- The enclosure still lacked proper mounting and weatherproofing
- Enclosure did not survive simulated rain testing (garden hose spray)



PCB layout drawn on the copper

Prototype V3 (Field-Ready)

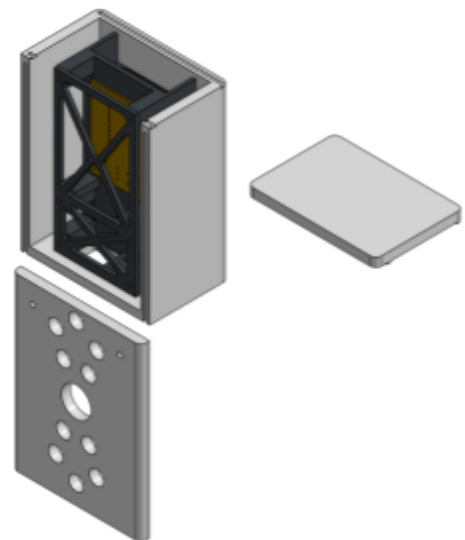
V3 addresses V2's limitations and represents a near-production design suitable for community deployment. This version features an improved weatherproof enclosure with a professional rain shield and particle filter.

V3 Enhancements:

- **Reference-based accuracy validation:** Co-located testing against EPA AQS station data, a borrowed PurpleAir sensor for PM2.5, and National Weather Service data for temperature/humidity/pressure provided an external benchmark for sensor accuracy.
- **User-friendly dashboard:** Color-coded AQI display, plain-language health recommendations, and mobile app with push notifications
- **Professional enclosure:** Outer protective enclosure for component housing. Ventilation vents positioned towards the bottom to prevent water and moisture from entering. Accessible sliding hatches on the outer enclosure to allow for inside access and mountability



Outer Enclosure



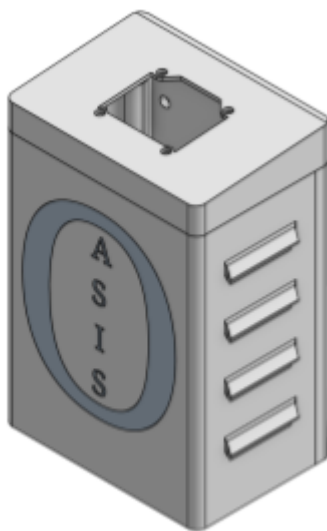
Sliding panels on enclosure

Prototype V4 (Current Version)

V4 builds upon V3's field-ready foundation, focusing on refining its size, serviceability, and internal organization in order to deliver a more compact, polished, and easily maintainable design suitable for widespread community deployment. V4 emphasizes a smaller physical footprint, improved water resistance, and a streamlined internal layout that makes the module both reliable and easy to troubleshoot.

V4 Enhancements:

- **Compact power system:** Switched from a bulky USB power bank to a compact 9V battery, reducing the module's overall size by roughly 20% while reducing battery runtime to about 3 hours. A voltage regulator steps the supply down to 5V to safely meet the ESP32's input requirements.
- **Side-mounted ventilation:** Relocated the vents from the bottom to a series of angled louvered openings on the side. Their sloped geometry promotes consistent airflow for accurate readings while simultaneously preventing electrical components from overheating, and causes water droplets to slide off rather than seep inside.
- **Snap-fit top cover:** Redesigned the top cover with a snap-fit mechanism, allowing for quick, tool-free access that simplifies maintenance and component inspection.
- **Improved internal electrical housing:** Reorganized the internal housing with dedicated wire management, reducing clutter, securing connections, and improving overall reliability and durability of electronics.



V4 Outer Enclosure



V4 Enclosure Outside

Data Visualization Interface:

Phase 1: Grafana (Placeholder)

To begin interpreting sensor data early in the project, the team deployed Grafana as an initial placeholder dashboard. Its primary advantage was speed and accessibility because getting data on screen required minimal custom development, and the platform offered a wide range of pre-built visualization options that could be connected to the sensor pipeline relatively quickly. For a team focused on validating hardware and data collection, it provided a convenient way to confirm that readings were coming through correctly and that the system was functioning as intended.

However, Grafana quickly revealed itself to be a poor fit as a long-term solution. While the platform is powerful in the right context, making meaningful use of it demands a significant investment of time and technical effort. Configuring panels required familiarity with query languages and data source syntax, and building dashboards that accurately reflected the environmental data proved more complex than anticipated. Beyond setup, the day-to-day experience of interpreting the data was long and arduous. The interface, while flexible, was not designed with accessibility or non-technical users in mind. Extracting actionable insight from the visualizations required constant manual effort, and the gap between what the data showed and what it actually meant in practical terms was difficult to bridge within Grafana's framework. It served its purpose as a proof of data flow, but it was clear that a purpose-built solution was needed to make the information genuinely useful.



Display of the readings from the PM2.5 sensor on the dashboard

To the left is a picture of the Grafana data visualization. Grafana is an open-source analytics and visualization platform that connects to data sources, including time series databases to display metrics, logs, and traces in customizable dashboards. It's primarily used for monitoring application performance and metrics in real time.

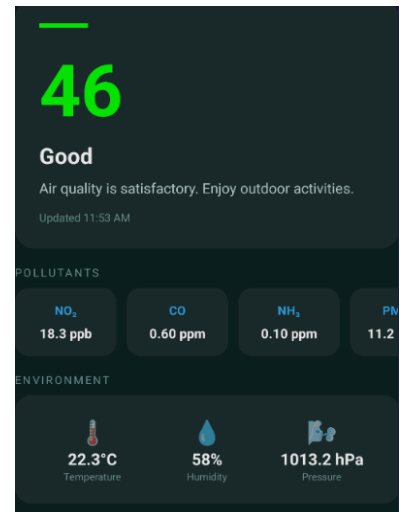
Phase 2: O.A.S.I.S. Mobile Application

The O.A.S.I.S. mobile application translates raw sensor readings into clear, usable information for community members, researchers, and responders. The app connects to the O.A.S.I.S. data pipeline through InfluxDB, allowing sensor readings from each node to be displayed in real time.

Home Dashboard

What it does: Displays current AQI, pollutant readings, environmental conditions, node ID, last update time, and health guidance.

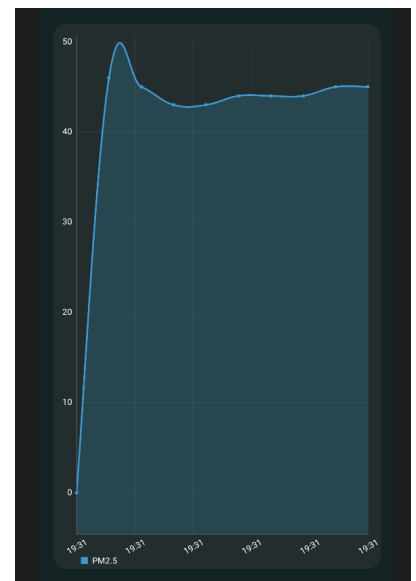
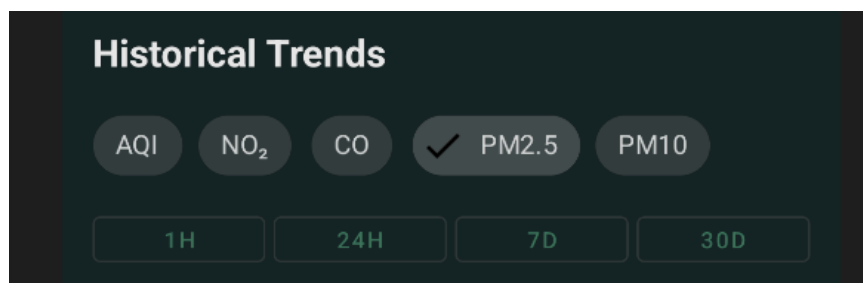
Purpose: Gives users an immediate understanding of current air quality without needing to interpret raw sensor data.



Trends Tab

What it does: Shows historical air-quality patterns across time ranges such as 1 hour, 24 hours, 7 days, and 30 days.

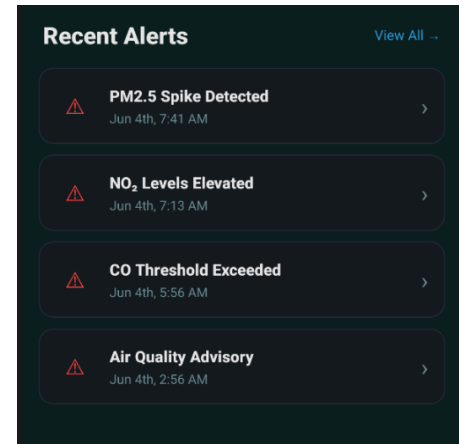
Purpose: Helps users identify pollution spikes, recurring patterns, and long-term environmental changes.



Alerts System

What it does: Records threshold-based advisories, health alerts, and emergency-level warnings when readings reach unsafe levels.

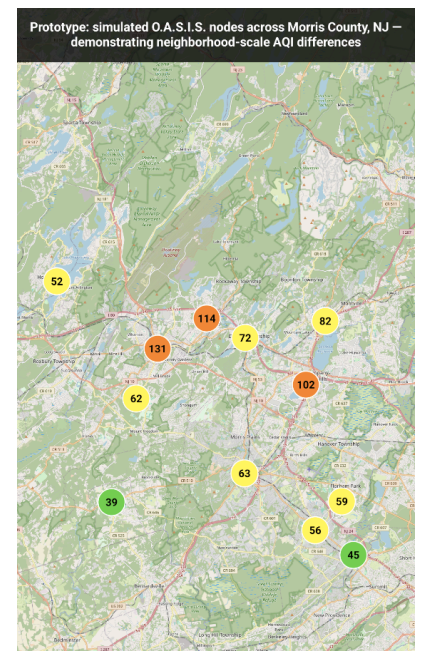
Purpose: Converts sensor data into actionable warnings so users can respond quickly to hazardous conditions.



Map Tab

What it does: Displays O.A.S.I.S. nodes as color-coded AQI markers across a community using prototype or simulated node data.

Purpose: Demonstrates how deployed nodes would reveal neighborhood-scale air-quality differences.



Profile and Data Export

What it does: Provides basic personalized guidance and exports timestamped sensor readings as CSV files.

Purpose: Supports individual health decisions while making the data usable for schools, labs, and citizen-science research.

Health Profile

Age Group

Adult (18–64) ▼

Health Conditions

☐ Asthma

☐ COPD / Cardiovascular Disease

☐ Regularly active outdoors

UPDATE ASSESSMENT

EXPORT DATA

Download sensor history as a CSV file.

Range (hours): 24

EXPORT CSV

YOUR RISK ASSESSMENT

Current AQI: 0 – Good

air quality is good. Enjoy outdoor activities safely.

Iteration Process

I. Narrative Testing

During testing, our prototype was functional but lacked some of the finalized hardware components and weatherproofing features needed for long-term outdoor deployment, so we broke down O.A.S.I.S.'s testability, potential refinements, effectiveness, and issues found during the iteration process:

Since a full-scale deployment of O.A.S.I.S. across an entire community network requires significant resources and infrastructure, we conducted a phased testing approach combining trials and comparison studies to evaluate system performance:

1. Prototype Validation Testing:

- Deployed the prototype unit in distinct environments around our local area:
 - **Location A (Roadside):** Near a busy intersection to capture vehicle emissions and traffic-related pollutants during times of peak rush hour traffic.
 - **Location B (Residential):** In a backyard setting to establish baseline air quality in a typical suburban environment.
 - **Location C (School Campus):** Next to the school building to monitor air quality where students spend their time and compare indoor vs. outdoor readings.
- Ran numerous 3-hour monitoring cycles to collect data and establish a baseline for readings
- Compared O.A.S.I.S. readings against:
 - **Government Station Data:** Downloaded publicly available data from the nearest EPA monitoring station, located several miles from our testing area, to assess correlation and identify local variations our system could detect.
 - **Commercial Monitors:** Borrowed a PurpleAir sensor from our science teacher to cross-reference PM2.5 readings and verify our sensor accuracy.
 - **Weather Station Data:** Used local weather service data to validate our BME280 temperature, humidity, and pressure readings.
- Documented key performance indicators:
 - Sensor consistency (comparing readings from data on different days, but same location and time)
 - Environmental correlation (how temperature and humidity affected pollutant readings)
 - Data upload success rate (percentage of readings successfully transmitted to InfluxDB)
 - Battery life under different weather conditions
 - Response time to pollution events (tested by monitoring readings during a nearby lawn mowing session and fireplace smoke)

2. Stress Testing and Durability Assessment:

- Subjected prototype enclosures to simulated weather conditions:
 - Sprayed units with mist to test water resistance
 - Left units in direct sunlight during peak afternoon heat to assess temperature tolerance
 - Monitored performance in freezer with freezing temperatures
 - Tested sensor response recovery after condensation from morning dew
- Evaluated power system performance:
 - Measured daily power consumption patterns
 - Tested battery backup duration during simulated power loss
 - Calculated charging efficiency on cloudy vs. sunny days for optional solar charging
- Conducted network reliability testing:
 - Monitored Wi-Fi connection stability at various distances from router
 - Tested backup connectivity in areas with weak Wi-Fi signal
 - Measured data transmission delays under different network loads



3. User Interface and Alert System Testing:

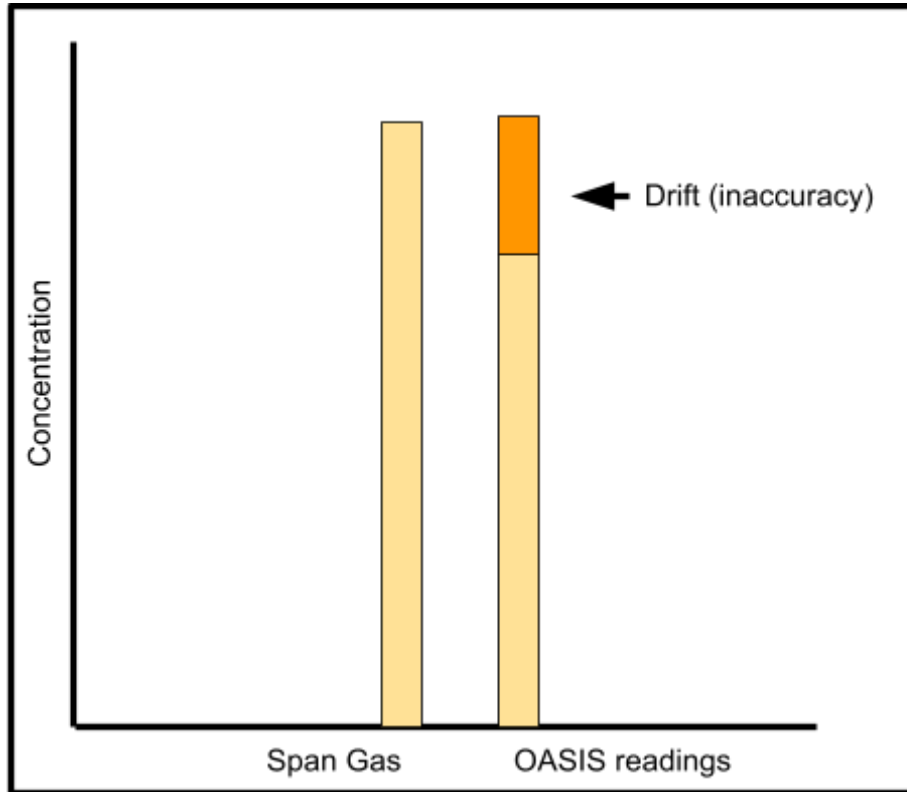
- Created a test dashboard accessible to our project team and several volunteer community members (parents, teachers, neighbors).
- Generated test alerts by manually triggering threshold conditions in the database to evaluate:
 - Alert delivery speed (how quickly users received notifications)
 - Message clarity (whether users understood the severity and recommended actions)
 - Dashboard usability (whether non-technical users could navigate and interpret data visualizations)
- Collected feedback through crowdsourcing (surveys asking testers):
 - Rate ease of understanding air quality readings
 - Identify confusing elements in the interface
 - Suggest improvements for data presentation
 - Evaluate whether urgency is displayed through alerts

II. Possible Refinements

Upon completing our current prototype of O.A.S.I.S., these were the refinements we identified regarding issues we would like to improve or refine given a larger time span and additional resources:

1. Sensor Accuracy and Calibration Enhancement

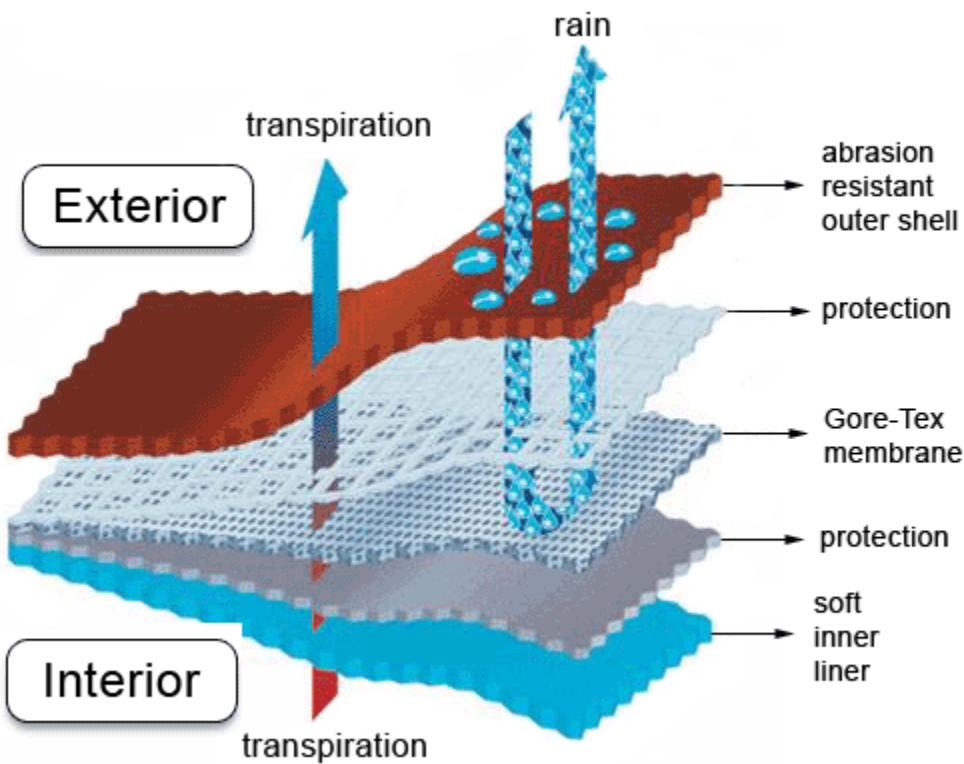
a. One of the limitations of our current prototype is that we were unable to afford research-grade sensors for pollutants such as NO_2 , NH_3 , and CO . Our current sensors are consumer-grade metal oxide modules that provide reasonable accuracy but may drift over time or show cross-sensitivity to multiple gases. We aspire to implement the repetitive calibration protocols seen in higher-precision sensors to prevent drifting accuracy. We would implement this change by partnering with a local university environmental science department or air quality research lab that could provide access to reference-grade equipment for comparison testing. Additionally, we would develop a calibration schedule where sensors are periodically exposed to clean air (pollutant-free) and span gas (known concentration) to improve accuracy and correct for drift. We'd also integrate temperature and humidity compensation algorithms into our data processing, since these environmental factors significantly affect metal oxide sensor performance. While this would require additional programming and testing time, it's crucial for ensuring O.A.S.I.S. provides reliable data that communities can trust for a multitude of decisions.



b. **Testing means:** Use a gas dilution system with certified calibration gases or clean air to expose sensors to known concentrations of NO₂, CO, and other pollutants. Compare sensor readings against reference instruments and calculate accuracy, precision, and detection limits. Conduct long-term drift studies by leaving sensors running continuously for 30+ days and measuring deviation from baseline.

2. Weatherproofing and Durability Improvements

a. Our current enclosure is 3D-printed in PETG, chosen for its UV resistance and outdoor durability over PLA. However, the FDM-printed shell still has limitations for long-term deployment because layer seams can allow gradual moisture entry over many rain cycles, and even UV-stable filaments can slowly embrittle after months of continuous sun exposure, risking cracking at mounting points. We would design a next-generation weatherproof housing using injection-molded ABS plastic or aluminum with a Gore-Tex membrane that allows air exchange while blocking water more reliably than printed seams can.



b. **Testing means:** Conduct accelerated weathering tests using heat guns and thermal cycling chambers. Perform IP (Ingress Protection) rating tests by subjecting enclosures to water spray and dust exposure. Monitor internal temperature and humidity over 24-hour cycles to ensure electronics remain within operating ranges.

3. Data Platform and User Interface Enhancement

a. Our current data visualization consists of a basic dashboard that displays real-time readings and simple line graphs with threshold-based alerts and basic trend graphs. We would expand this into a comprehensive community air quality platform with interactive mapping, historical trend analysis, and predictive alerts. The enhanced platform would include color-coded air quality heat maps showing pollution across multiple monitoring stations, allowing users to see which neighborhoods have better or worse air quality. We would implement machine learning algorithms that analyze patterns in the data to predict pollution events based on time of day, weather conditions, and historical trends, for example, alerting users that air quality typically degrades during evening rush hour on humid days. A machine learning algorithm acting on more data would provide better output and improved data interpretation. The mobile app would include personalized health recommendations based on user profiles (age, respiratory conditions, activity level) and current air quality readings, such as "Air quality is moderate. Sensitive groups should reduce prolonged outdoor activity." We would also add a community reporting feature where users can submit observations about visible smoke, industrial emissions, or unusual odors that might correlate with sensor data spikes. Finally, we would implement data export tools that allow researchers and policy advocates to download historical datasets in CSV format for independent analysis.

b. **Testing means:** Conduct additional crowdsourcing with diverse community members representing different age groups and technical literacy levels. Track user engagement metrics including dashboard views, alert response rates, and feature utilization. Adjust data sorting algorithm to handle data from 100+ sensors and hundreds of simultaneous users without performance degradation and perform load testing.

4. Network Connectivity and Data Transmission Optimization

a. Our prototype relies on Wi-Fi connectivity, which limits deployment to locations near wireless networks and creates data gaps when connectivity fails. We would implement a dual-mode communication system with Wi-Fi as the primary connection and backup data on an SD card for manual recovery. This would allow O.A.S.I.S. stations to be deployed in remote areas like forests or rural communities without reliable Wi-Fi coverage. We would also optimize our data transmission protocol by implementing intelligent buffering-if network connection is lost, the system stores readings locally on an SD card and automatically uploads the backlog once connectivity is restored, ensuring no data loss. To reduce cellular data costs for stations using mobile networks, we would compress data packets and implement adaptive transmission intervals that send updates every 5 minutes during normal conditions but increase to every 1 minute during pollution events.

Additionally, we would implement mesh networking using LoRaWAN, which extends communication range beyond Wi-Fi. LoraWan enables several kilometers per hop versus roughly 100 feet for Wi-Fi and allows nearby O.A.S.I.S. units to relay data through each other to a gateway node. This would let stations deployed beyond Wi-Fi range, such as rural areas or remote roadside locations, transmit data through

neighboring nodes rather than requiring direct internet access at every unit, which would provide the network-wide architecture the system was designed around.

b. **Testing means:** Measure data transmission success rates under various network conditions including weak Wi-Fi signals, cellular dead zones, and simulated network outages. Calculate data usage per day for cellular connections and verify compression algorithms maintain data integrity. Test mesh network relay functionality with multiple units at varying distances to determine effective range and reliability.

III. Reflection Over Effectiveness

O.A.S.I.S. directly addresses the challenge of hyperlocal air quality monitoring by providing communities with real-time, granular pollution data that traditional government monitoring stations cannot offer due to their sparse geographic distribution. Shifting away from the single-station, city-wide average model, O.A.S.I.S. uses an ESP32 firmware stack to stream data continuously to an InfluxDB pipeline visualized through our custom Android app, allowing residents to observe neighborhood-level variations that aggregate readings would otherwise mask. A future AI-driven anomaly detection layer is planned to flag pollution spikes in real time, which would give communities additional lead time to protect vulnerable populations, building on the threshold-based alert system currently implemented.

1. Sensor Accuracy and Calibration Validation

- **Strength:** Co-locating O.A.S.I.S. units alongside certified reference instruments at an EPA or university air quality monitoring site would provide a direct, quantitative benchmark for sensor accuracy across all measured pollutants. Cross-referencing against a PurpleAir sensor for PM_{2.5} and local weather service data for BME280 readings adds additional verification layers. Deploying units across environmentally distinct locations (roadside, residential, school campus) simultaneously tests whether the system resolves spatial variation that single reference stations miss, which is the core value proposition of the network model.
- **Weakness:** Consumer-grade metal-oxide sensors on the MICS6814 are susceptible to cross-sensitivity between gases and long-term drift, meaning short-term accuracy tests may not reflect multi-month field performance. Without access to a gas dilution system with certified calibration gases, establishing true detection limits and precision for NO₂, CO, and NH₃ channels remains difficult.
- **Proof of Plausibility:** If O.A.S.I.S. readings fall within an acceptable margin of reference instrument values across multiple pollutants and environmental conditions, and if the system detects localized pollution events (such as vehicle emissions or combustion sources) that the distant reference station cannot resolve, this validates that the sensor suite provides reliable, meaningful data at the community level.

2. Durability and Long-Term Deployment Testing

- **Strength:** Subjecting the enclosure to accelerated weathering protocols covering sustained water exposure, UV radiation, peak heat, near-freezing temperatures, and condensation cycling would

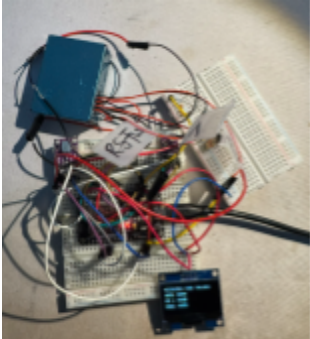
directly probe the failure modes most likely to limit outdoor deployment lifetime. Pairing this with continuous power system monitoring would evaluate whether the AC-powered design can maintain stable operation during extended testing, temporary outages, and continuous data transmission.

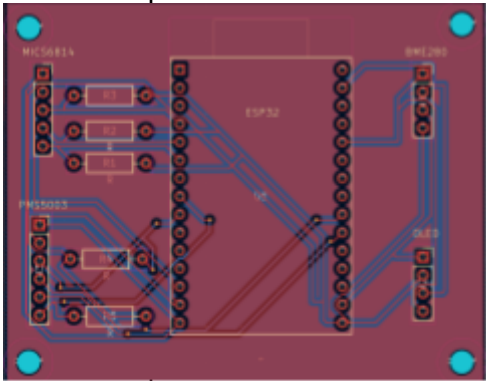
- **Weakness:** Accelerated lab weathering is an approximation of real multi-season exposure and may not capture cumulative effects like UV embrittlement of the PETG enclosure, gradual seal degradation, or particulate fouling of sensor intake vents over months of continuous operation. True durability validation would require extended outdoor deployment across full seasonal cycles.
- **Proof of Plausibility:** If the enclosure maintains sensor operability, data transmission continuity, and power sufficiency across the full range of simulated environmental stresses, this supports the claim that O.A.S.I.S. can sustain the uninterrupted outdoor deployment necessary for longitudinal community air quality monitoring without frequent maintenance intervention.

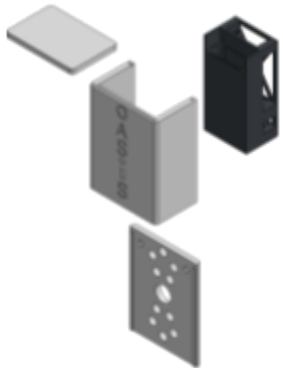
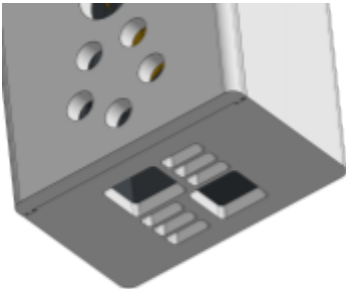
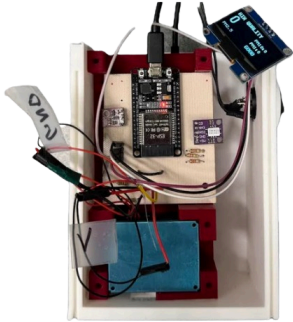
3. Community Interface and Alert System Evaluation

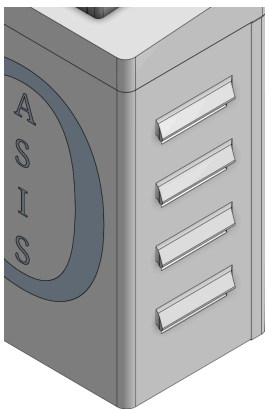
- **Strength:** Structured usability testing with community members spanning diverse age groups and technical literacy levels would directly evaluate whether the custom mobile app and alert system serve their intended non-specialist audience. Generating controlled alert conditions by manually triggering threshold violations in the database allows measurement of alert delivery speed, message clarity, and appropriate user response, all of which determine whether the system translates raw sensor data into real protective action.
- **Weakness:** A volunteer test pool drawn from familiar social networks may skew toward users already motivated to engage with air quality data, limiting the generalizability of usability findings. Load testing the platform against a large simultaneous user base and a dense multi-station network would be necessary to validate scalability before any wide community deployment.
- **Proof of Plausibility:** If community testers across varying backgrounds can correctly interpret AQI readings, act appropriately on threshold alerts, and navigate historical trend data without technical assistance, this demonstrates that O.A.S.I.S. functions as a practical public-facing tool rather than a research instrument, fulfilling the system's core mission of democratizing environmental data for communities that have historically lacked access to it.

IV. Completed Refinements and Issues

Version	Aspect	Reasoning
V1	Microcontroller Selection	Once V1 confirmed our core sensor and data pipeline concept, we recognized the Arduino Uno would limit further development because it lacks native Wi-Fi or BLE, requiring an additional shield to transmit data to InfluxDB, and has limited I/O for our three-sensor array. We transitioned to the ESP32, which integrates Wi-Fi and Bluetooth directly on the chip, offers a dual-core processor for handling sensor polling and network transmission simultaneously, and provides enough GPIO/ADC channels for our full sensor suite without added hardware. This consolidation also reduced the component count and wiring complexity carried into the V2 PCB design.
	Compact Breadboardability  Inefficeint breadboard circuit	We initially intended to use a breadboard in our final design, as it allowed for easy changes and fixes to our circuit if needed. However, we realized if we wanted to maintain our criteria of designing a compact system, then our circuit would have to be much smaller than a breadboard, especially given that we were considering utilizing two at one point. Thus, we started moving towards a printed circuit board.
	On/Off switch	Once deployed, O.A.S.I.S. is designed to draw continuous power directly from the home's electrical supply, allowing it to monitor and record atmospheric conditions around the clock without interruption. Because the module's core purpose is to capture air quality data at all times, uninterrupted operation is essential to its function. For demonstration and easy testing purposes, however, we equipped the prototype with a 9V battery and an integrated on/off rocker switch.
	System Chassis	Our initial V1 prototype used simple press fits for our breadboard, sensors, and battery. However, this chassis had serious limitations, we designed the plastic too thin, and it cracked easily when drilling mounting holes. Furthermore, the ventilation holes allowed direct sunlight to heat the interior, causing temperature sensor readings to be inaccurate. We attempted to solve the heating problem by covering the holes with mesh, but this created an unprofessional appearance. The lack of proper ventilation ports meant we had to leave the chassis slightly ajar for airflow, which completely defeated weatherproofing.

V2	OLED Screen 	For V2, we added a small OLED display that showed real-time sensor readings directly on the unit, thinking this would provide immediate visual feedback and make the system more user-friendly during field deployment. The screen displayed current PM2.5, temperature, humidity, and AQI values that updated every 10 seconds, giving us a convenient way to verify sensor operation without needing to access the web dashboard. During initial testing in our lab, the OLED proved incredibly useful for troubleshooting—we could instantly see if sensors were working and whether data looked reasonable. The display added a professional touch to our prototype. However, we discovered that in actual outdoor deployment, the small screen was difficult to read in direct sunlight due to glare, and the OLED's power consumption (20-30 mA continuously) reduced our battery runtime noticeably. We also realized the feature was somewhat redundant since users could easily check air quality on their phones through our mobile app, which provided much more detailed information and historical trends.
	Grafana database 	We initially implemented Grafana as our data visualization platform because it offered professional dashboards with minimal coding required. However, as we prepared for actual community deployment, Grafana's limitations became apparent. The platform required users to understand its complex interface that overwhelmed non-technical community members during user testing. Most importantly, Grafana lacked the mobile-friendly interface and personalized alert features that our community users needed. We ultimately transitioned to InfluxDB for data storage combined with a custom-built web interface and mobile app that we could tailor specifically to community air quality monitoring needs, providing simplified views and plain-language health recommendations.
	Custom Printed Circuit Board (PCB) 	Transitioning from breadboard prototyping to a custom PCB was one of our most significant V2 improvements. We designed the PCB using KiCad software, carefully routing traces to minimize interference and positioning components to reduce the overall board footprint by 60%. Instead of ordering professionally manufactured boards, we used the hand-etching method with copper-clad boards, transferring our printed circuit design using the toner transfer technique and etching with ferric chloride solution. This DIY approach allowed us to produce PCBs in our school lab within a day rather than waiting weeks for manufacturer delivery. However, hand-etching had limitations: achieving fine trace spacing was difficult, and the etching process sometimes produced broken traces that required manual fixes with solder. Drilling accurate holes for components was time-consuming and prone to misalignment. Despite these challenges, hand-etched PCBs eliminated the jumper wire mess and gave us valuable hands-on experience with PCB fabrication while keeping costs under \$5 per board.
	Accessible Housing Enclosure	For V2, we upgraded from our makeshift food container to a custom 3D-printed enclosure that we designed specifically for O.A.S.I.S.'s needs. We used PLA filament and refined the design in CAD software to create integrated mounting

	 Designed sliding walls for easy access to the inside	points for sensors, cable routing channels, and snap-fit assembly features that made maintenance easy. The 3D-printed design allowed us to rapidly prototype different enclosure shapes without expensive tooling costs, and we could customize the size precisely to fit our PCB and sensor array. We incorporated recessed areas for the BME280 sensor to shield it from direct sunlight while maintaining airflow, and designed built-in standoffs that secured the circuit board without requiring separate hardware. The ability to iterate designs quickly was invaluable-when we discovered our component spacing needed adjustment, we simply modified the CAD file and printed a new version overnight. We switched to PETG filament for better outdoor durability and added UV-resistant coating to prevent degradation from sunlight exposure.
V3	System Ventilation  Bottom vent holes	Achieving proper ventilation while maintaining weatherproofing was one of V3's most challenging refinements. Our V2 enclosure's simple drilled holes allowed insects, dust, and water to enter during heavy rain. We researched professional weather station designs and designed vents at orientations that allow air molecules to pass through while preventing water droplets. We installed two vents positioned at the bottom of the module to create natural convective airflow across sensors. Field testing simulation showed this ventilation system maintained sensor accuracy while keeping internal electronics completely dry through multiple rainstorms.
	OLED stayed	We retained the OLED display despite V2's noted limitations, since it lets technicians verify sensor and network status on-site during installation and maintenance without needing a paired phone. The mobile app remains the primary interface for end users.
	Improving Etching Techniques 	While V2's hand-etched PCB was a major improvement over breadboards, it was still designed with conservative component spacing due to our DIY etching limitations. For V3, we redesigned the PCB with tighter layouts and an improved laser etching technique, reducing the board area by 30-40%. We refined our process by taping the copper-clad board, laser engraving the trace pattern, applying Sharpie to protect the exposed copper, and then etching in acid. This allowed us to achieve much finer traces. This compact design allowed us to shrink the enclosure, reducing 3D printing material costs. However, the denser layout made etching more challenging. We required several prototypes to optimize space between traces to prevent bridges from overlapping. We also had to ensure the ESP32's Wi-Fi antenna had adequate clearance from other traces to prevent interference. Despite these complications, our improved laser etching process created a professional-looking monitoring station while maintaining rapid in-house prototyping capabilities.
	USB Power bank	For O.A.S.I.S.'s portable power source, we selected a USB power bank, which

		offered stable, reliable output along with excellent capacity for extended runtime. A key advantage of the USB power bank was that its 5V output directly matched the ESP32's input requirements, allowing us to power the microcontroller without the need for an additional voltage regulator or step-down circuitry. This not only simplified our wiring and reduced the number of components within the enclosure, but also provided a clean, stable voltage supply that protected the board from potential damage, making the power bank an efficient and dependable choice for continuous sensor operation.
	App development	Creating a user-friendly mobile application was critical for making O.A.S.I.S. accessible to non-technical community members. We developed the app using Kotlin and Java in Android Studio. The app displays the current Air Quality Index with a large color-coded indicator that's immediately understandable without technical knowledge. During beta testing with local community members, we discovered that elderly users struggled with complex navigation, leading us to simplify the interface to just a few main tabs and additional images for context. The biggest development challenge was efficiently handling real-time data updates without draining phone batteries. We implemented an algorithm that updates at specific intervals in short spurts to conserve power while not sacrificing temporal accuracy.
V4	Improved System Ventilation 	Unlike V3's bottom-oriented ventilation, V4 relocates the vents to a series of angled louvered openings on the side. Rather than simple holes, these slanted slats allow air to flow freely across the sensors while their downward-facing geometry blocks the direct entry of water, dust, and debris. Positioning the vents along the side wall promotes consistent airflow over the sensor array, drawing in fresh ambient air for accurate, real-time readings instead of measuring stagnant air trapped inside. This continuous exchange also dissipates heat from the electronics, preventing the sensors from overheating and skewing their measurements. When tested with water, the sloped louvers caused droplets to slide off rather than pool or seep inside, keeping the internal electronics completely dry while preserving full ventilation.
	More Compact Design	One of the biggest changes in V4 was reducing the overall size of the module by changing its power source. By switching from a bulky USB power bank to a compact 9V battery, we were able to make O.A.S.I.S. roughly 20% smaller, significantly reducing its physical footprint. This allows for a sleeker, more streamlined enclosure that is far less obtrusive and easier to mount, though this came at the cost of battery runtime. The reduction in size made O.A.S.I.S. more practical for everyday deployment without compromising its monitoring capabilities.
	9V Battery vs. Power Bank	When selecting a portable power source, we weighed the pros and cons of a USB power bank against a 9V battery. The power bank offered stable output and

		excellent capacity, but its bulky form factor took up significant space within the enclosure. The 9V battery was much more compact. Its only tradeoff was voltage compatibility, since the 9V supply exceeds the ESP32's requirements, but this was fixed by adding a voltage regulator to step it down to 5V, providing a clean, stable input that protected the microcontroller while preserving valuable space. This comparison applies specifically to the portable prototype used for field testing and demonstration. The production design is intended to run on continuous home AC power rather than battery power.
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Communication of Solution

O.A.S.I.S. directly addresses the Grand Engineering Challenge theme of Engineering the Tools of Scientific Discovery by confronting one of environmental science's most persistent structural failures: the concentration of precision atmospheric instrumentation in well-resourced institutions, while rural, and underserved communities remain relatively unmonitored. As a result, many people living in these areas are unaware and unable to advocate for change or participate in scientific data analysis. O.A.S.I.S. dismantles this barrier by deploying **consumer-grade sensors cross-validated against reference instruments** to deliver data at a fraction of the cost, ensuring the tools of scientific discovery **reach those on the front lines of environmental hazard**.

Each O.A.S.I.S. module integrates a PMS5003 particulate sensor for PM2.5/PM10, a MICS6814 gas sensor for NO₂, CO, and NH₃, and a BME280 sensor for temperature, humidity, and pressure. These sensors connect to a custom PCB powered by an ESP32 microcontroller, resulting in **adequate atmospheric monitoring at approximately \$40**, a fraction of the cost of a professional unit

The companion mobile application allows students, researchers, citizen scientists, and emergency responders to monitor live conditions, receive hazard alerts, and export timestamped data. This makes participation in atmospheric research as accessible as checking a weather application

Timestamped data exportation capabilities in the form of CSV or JSON files allow anyone with an OASIS node to be able to share structured, collected data in detail to a nearby research laboratory, promoting citizen science. On the other hand, said research laboratory could deploy nodes in desired locations for themselves to collect data, assisting up-and-coming or minimally funded research groups

By placing modules directly into affected communities and through interconnected nodes, O.A.S.I.S. actively expands the frontier of environmental science. During wildfire events, during which PM2.5 can spike to 10–20 times safe thresholds, O.A.S.I.S. nodes capture this hyper-local atmospheric data that distant government stations may not capture immediately, and syncs it with nearby available nodes, producing structured records that could support research or real-time emergency response.

To summarize, O.A.S.I.S. addresses the Grand Challenge through several key mechanisms:

- **Consumer grade sensors and infrastructure** available off-the-shelf or funded by underserved communities at a fraction of the cost of other atmospheric instruments
- **An ease-of-use mobile application** allowing anyone, anywhere in the world to participate in collecting data, also allowing them to be able to configure hazardous alerts and constant updates on nearby atmospheric conditions.
- **Data exportation capabilities** promotes the accessibility of data to various research groups, and the collection of data through citizen science, further driving discoveries and breakthroughs
- **Interconnected deployment in under-resourced areas** breaks barriers and balances out instrument densities, allowing for synced atmospheric recording, regardless of the capability of an area.

Ultimately, O.A.S.I.S. exemplifies the spirit of engineering the tools of scientific discovery by reducing cost and complexity while preserving scientific usefulness for community-scale observation and expanding who can observe the world, contribute to its understanding, and participate in the ongoing process of discovery.

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
11/2/25 - 11/8/25 1.	Brainstorming	4 Hours	RR, RI, SH, AV	-Identified problem -Brainstormed possible solutions and researched on a google document
11/28/25 -11/30/25 2.	Identified three possible solutions & finalized on one	3 hour	RR, RI, SH, AV	-Further researched on the thre possible solutions identified - Determined which solution is best - Started/Updated portfolio
12/2/25 - 12/5/25 3.	Researched ways to deisgn circuit and 3D models	2.5 hour	RR, RI, SH, AV	- Researched microcontrollers & best sensors to use for the project - Made parts list for the project - Made mockup circuit
12/15/25 - 12/17/25 4.	Designed & programmed circuit	3 hours	RI, AV	- Breadboarded circut and programmed ESP32 to work with sensors
12/25/25 - 12/29/25 5.	3D modeled and printed system chasis	6 hours	RR, SH	- Modeled the chasis which breadboard would sit on
1/20/26 - 1/30/26 6.	Integrated displaying features (OLED, Grafana)	3 hours	SH, AV	- Integrated an OLED display into the circuit - Created means of dshowing data recorded on a Grafana dashboard

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1/25/26 - 1/31/26 1.	Created custom PCB	3 hours	RI, AV	- Drew, cut, and manufactured our first PCB according to our circuit
2/1/26 - 2/7/26 2.	3D modeled/printed outer and remodeled inner enclosures	6 hours	RR, SH	- Remodeled inner chassis according to the new PCB and other modifications - Created outer enclosure for the inner chassis, for greater protection
2/27/26 3.	Redesigned PCB for compactability	2 hours	RI, AV	-Redrew and recut PCB
2/28/26 4.	Assembled all parts and finalized product	2 hours	RR, RI, SH, AV	- Assembled all parts to make the final product
3/9/26 - 3/11/26 5.	Developed App & worked on portfolio components	3 hours	SH, AV	- Created android app to move away from Grafana for better user experience
3/12/26 - 3/19/26	Worked on portfolio components	4 hours	RR, RI, SH, AV	- Developed iteration process and chosen solution pages

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
3/23/26 - 3/28/26 1.	Continued to work on portfolio components	5 hours	RR, RI, SH, AV	-Worked on unfinished components within our portfolio
3/30/26 2.	Finalized portfolio	3 hours	RR, RI, SH, AV	-Final touchups on everything within the portfolio
4/17/26 3.	Made schedule for what to get done before National Conference	1 hour	RR, RI, SH, AV	Worked together to make a list of action items we needed to get done before nationals, and prioritized them based on how long our school was still open, or rubric from states, etc.
4/20/26 4.	Decided on what to change on trifold	1 hour	RR, SH, RI, AV	Looked through feedback and made a list of what to change on trifold
4/21/26 - 4/25/26 5.	Made changes to the trifold	3 hours	RI, SH, AV	-Made respective changes to our trifold, such as bigger text, more graphics, and evidence backed data -Reprinted it at our school
5/4/26 6.	Started updating the physical model	1 hours	RR, RI, SH, AV	Listed out changes we needed to do on our display items, and prototype itself

TECHNOLOGY STUDENT ASSOCIATION WORK LOG				
Date	Task	Time involved	Team member responsible (student initials)	Comments
5/6/26-5/11/26 1.	Updated the physical model	9 hours	RR, RI, AV	-Enhanced our prototype mount -Redesigned and reprinted prototype -Planned for more display elements, such as tablet stand, testers, etc.
5/18/26 2.	Started making plans to realize our app	2 hours	SH, AV	-Moving the app from being more concept to being used in the field Made a list of features we want, and how we would accomplish them
5/19/26 - 5/21/26 3.	Developed our apps front end	4 hours	SH, AV	-Worked on the displaying of data within the software -Worked on the UI
5/22/26 - 5/25/26 4.	Finalized our physical display materials	5 hours	RR, RI, AV, SH	-Pasted trifold onto posterboard -Created tablet stand for tablet to show app -Created a scenic display environment
6/1/26 - 6/4/26 5.	Worked on backend of the app	4 hours	AV, SH	-Added ways to export our data as a csv to allow for real lab analysis -Incorporated JSON -Added a real-time alerts feature

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
6/7/26 - 6/13/26 1.	Made a script for nationals and rehearsed	7 hours	AV, SH, RI	-Created a script for a nationals presentation -Rehearsed it every other day for about an hour
6/12/26 - 6/17/26 2.	Made final changes to portfolio	7 hours	RR, AV, RI, SH	-Reflected on comments on our rubric from states and made the respective corrections, updates, and changes -Also rehearsed script during this time
6/20/26 - 6/21/26 3.	Packed for nationals	3 hours	RR, AV, RI, SH	-Got everything ready for transportation, portfolios in USBs, and mentally prepared for the trip
4.				
5.				
6.				

STUDENT COPYRIGHT CHECKLIST (for students to complete and advisors to verify)

STUDENT: Answer question 1 below.

- 1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☒ NO

If NO, go to question 2.

If YES, is the music and/or sound copyrighted? ☐ YES ☐ NO

If YES, move to question 1A. If NO, move to question 1B.

1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.

1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, SEAN TISH (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

- 2) Does your solution to the competitive event integrate any graphics/videos? ☒ YES ☐ NO

If NO, go to question 3.

If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☒ NO

If YES, move to question 2A. If NO, move to question 2B.

2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.

2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, SEAN TISH (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

- 3) Does your solution to the competitive event use another's thoughts or research? ☒ YES ☐ NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? ☒ YES ☐ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, SEAN TISH (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Initials: AV RR RI SH

Chapter Advisor Signature: SEAN TISH

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